Lesson 05 Demo 04

Fetching Data Using the useReducer Hook

Objective: To demonstrate how to fetch data in a React application using the useReducer Hook, leveraging Vite for a faster and optimized development experience

Tools required: Visual Studio Code and Node.js

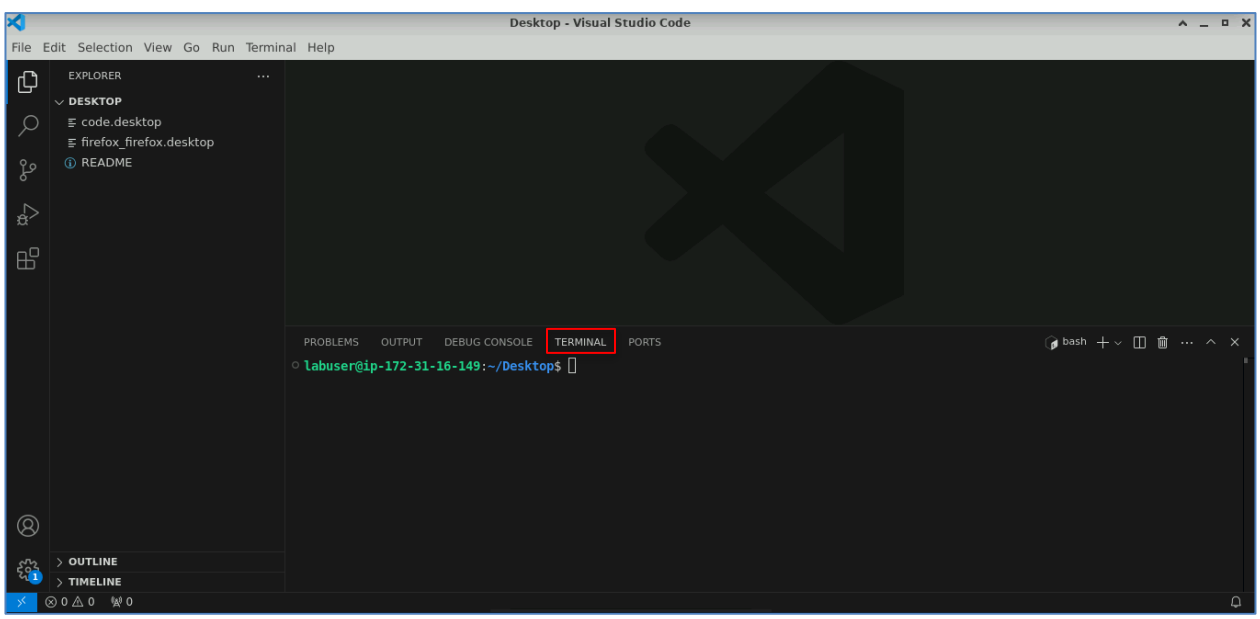
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Modify the App.jsx file
3. Run and verify the application

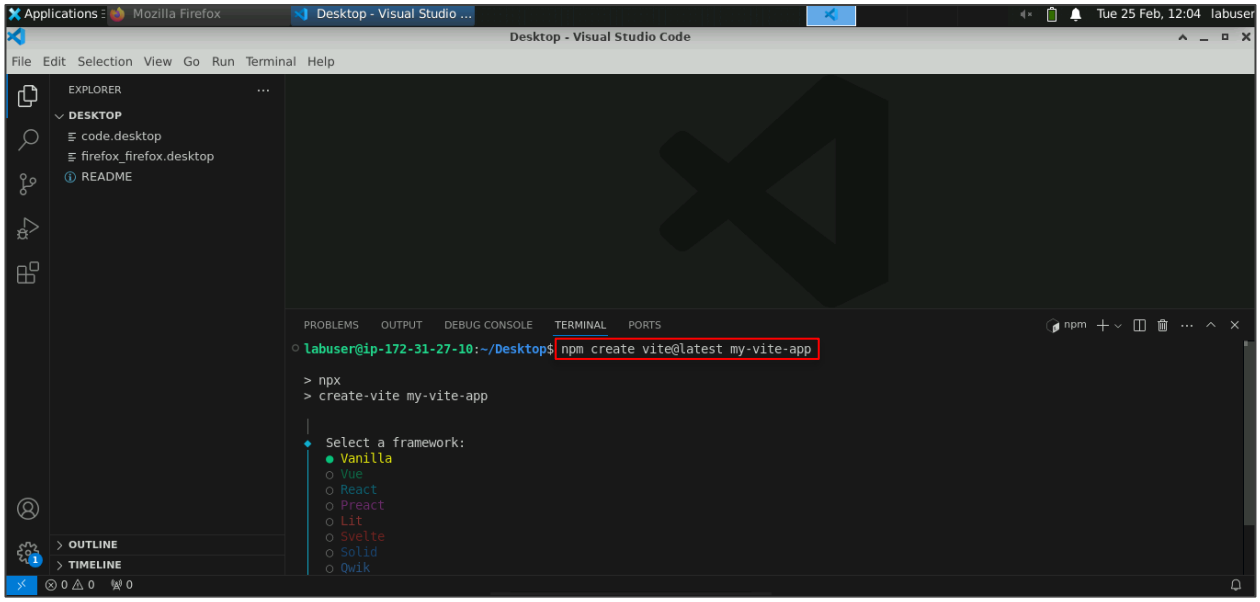
Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code (VS Code) and launch the terminal

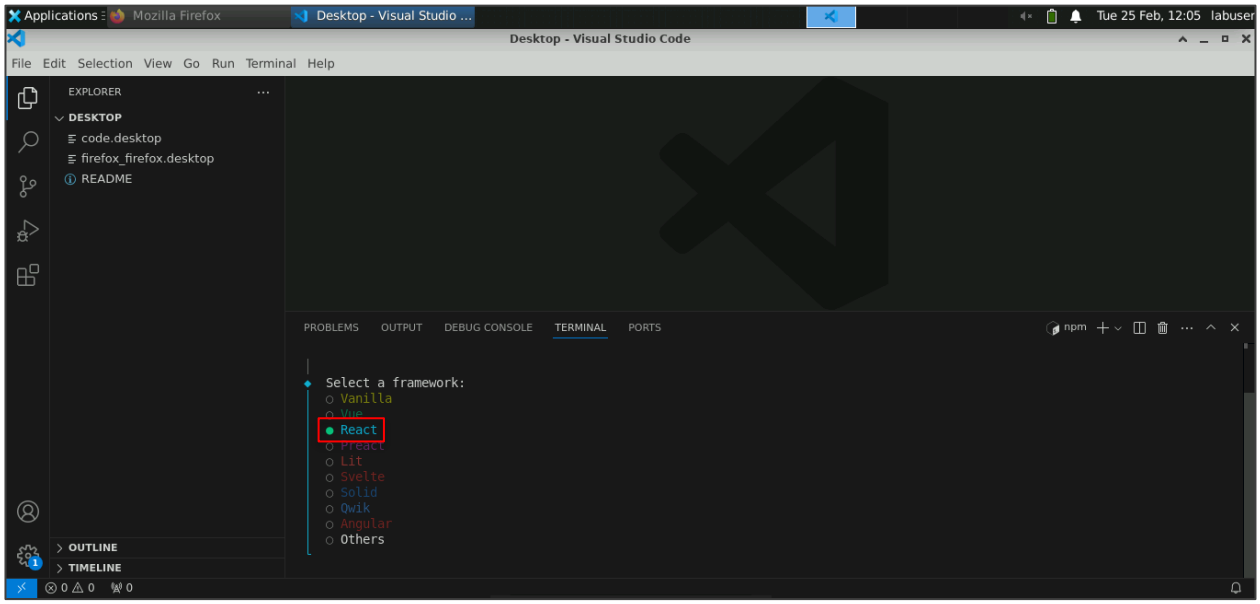


1.2 Run the following command to create a new React project using Vite:

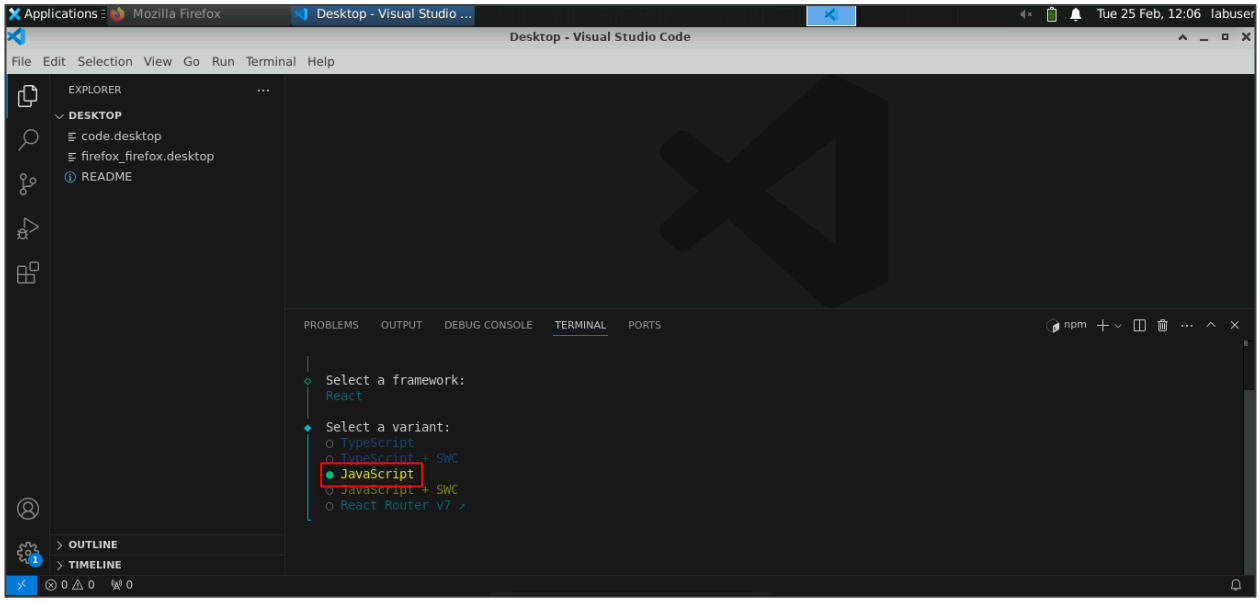
npm create vite@latest my-vite-app



1.3 Select **React** as the project framework and press enter

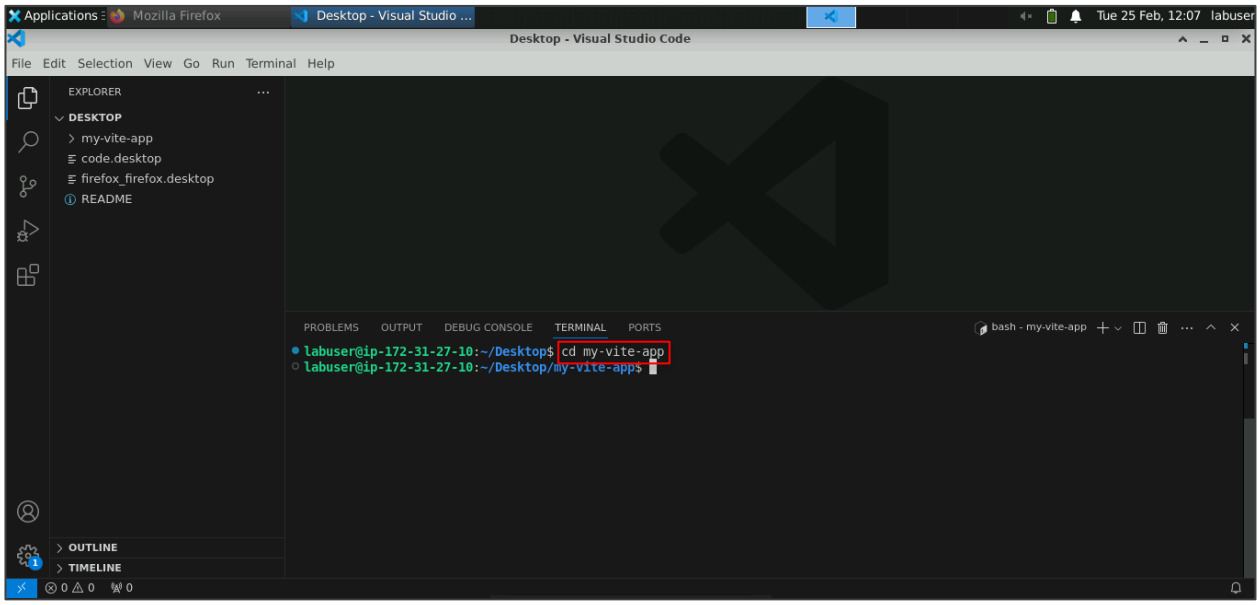


1.4 Select **JavaScript** as the language and press enter



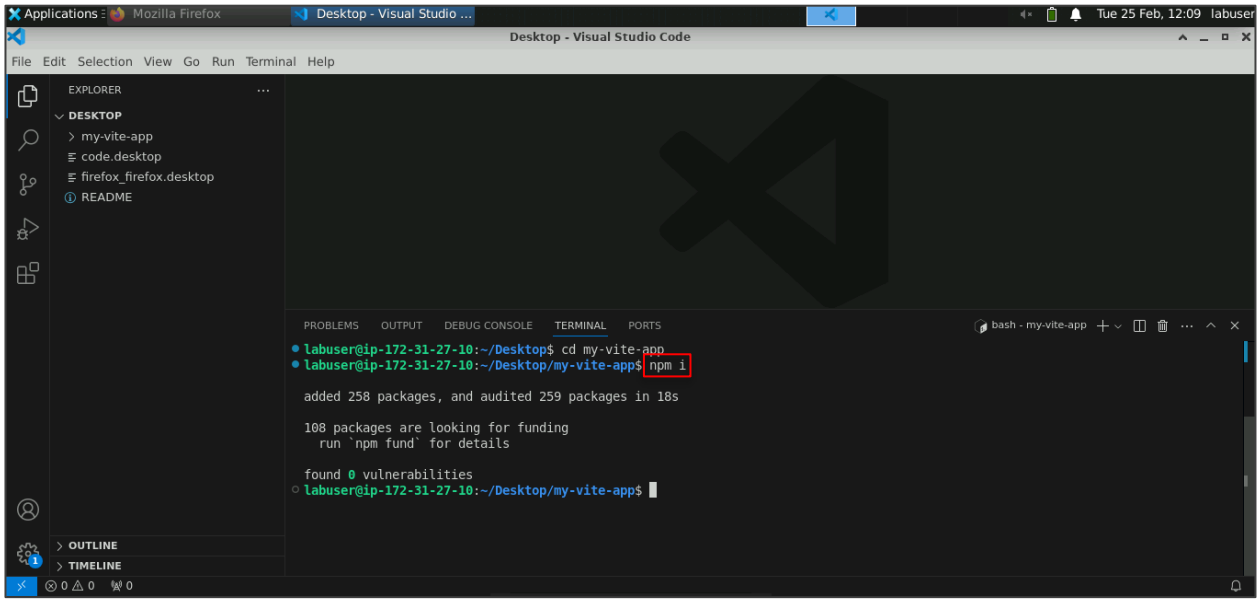
1.5 Run the following command to navigate into the project folder:

cd my-vite-app



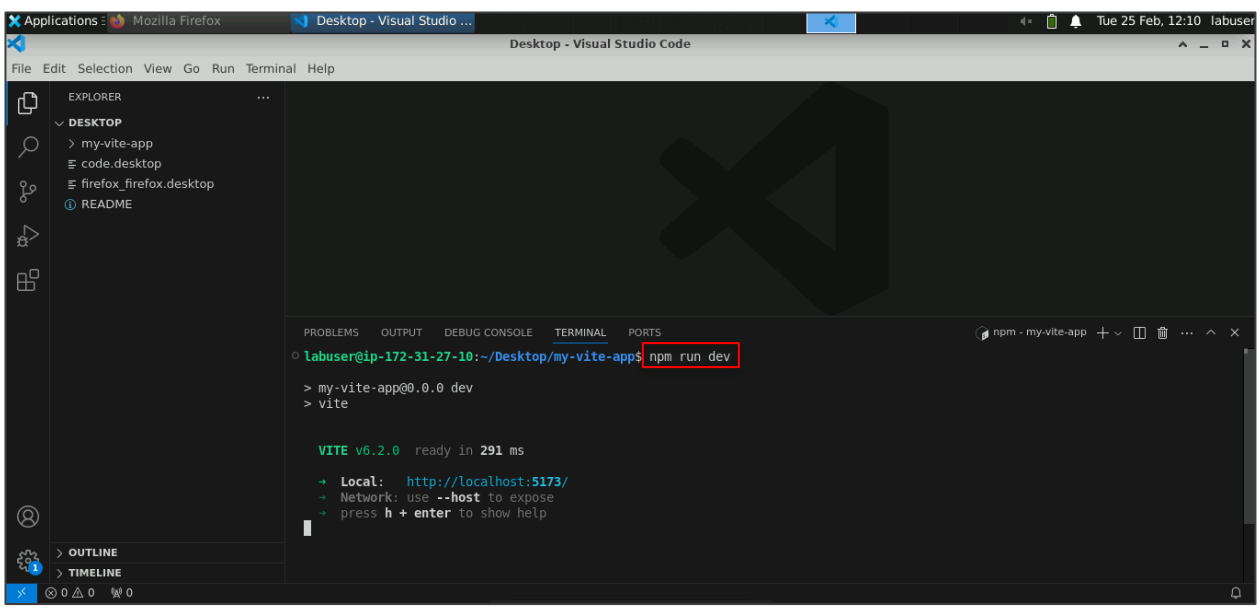
1.6 Run the following command to install dependencies:

npm i



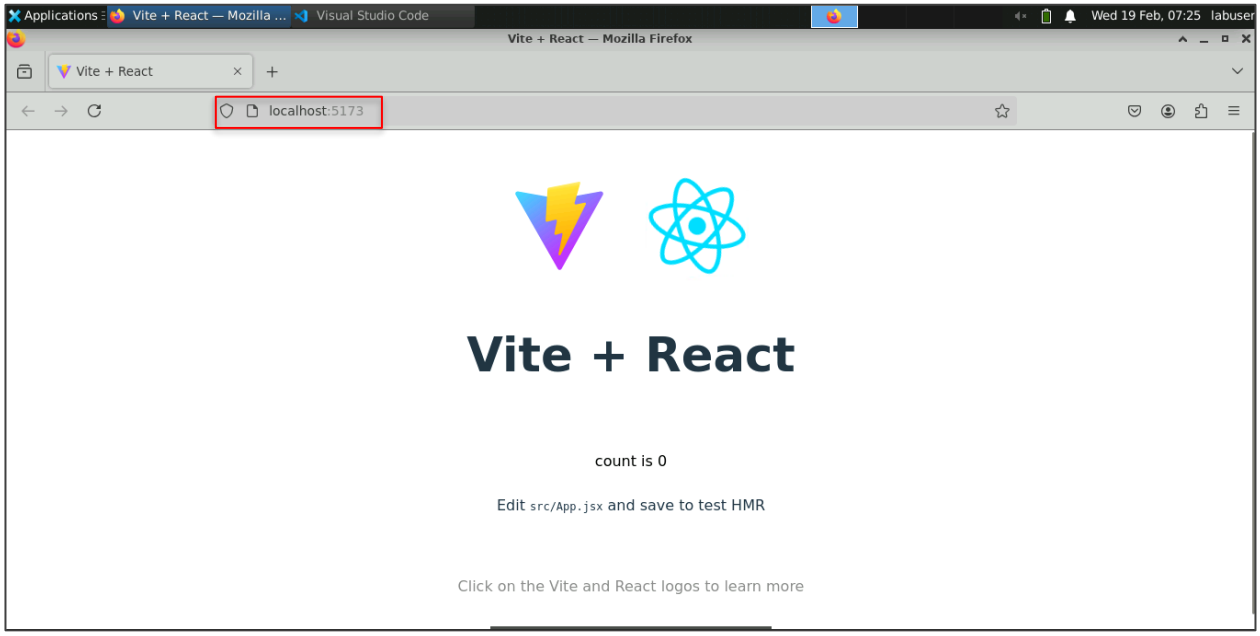
1.7 Run the following command to start the React development server:

npm run dev



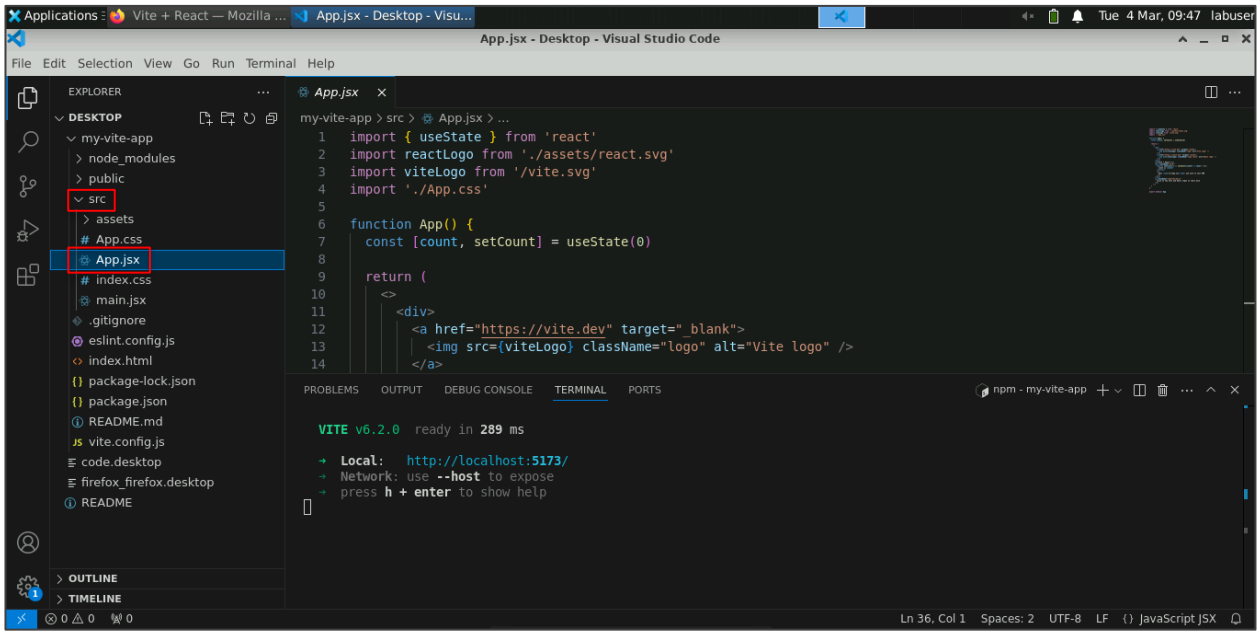
1.8 Navigate to the web browser and paste the following link:

http://localhost:5173



Step 2: Modify the App.jsx file

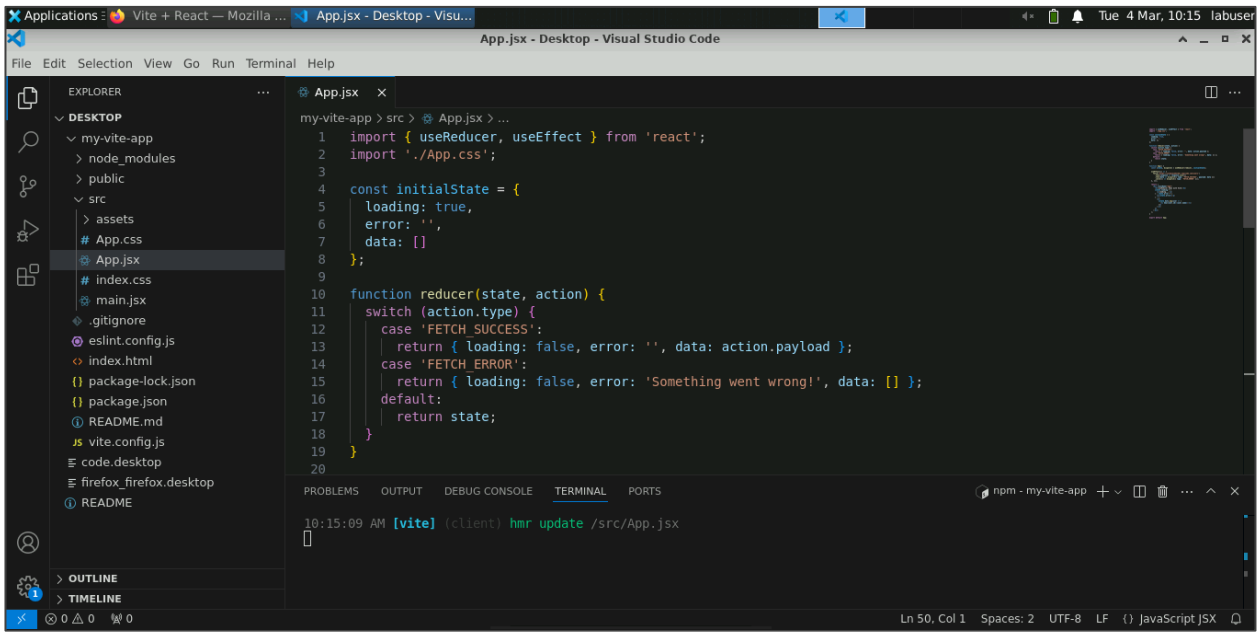
2.1 Open the **App.jsx** file inside the **src** folder



2.2 Paste the following code to include the new components in **App.jsx**:

```
import { useReducer, useEffect } from 'react';
import './App.css';

const initialState = {
  loading: true,
  error: '',
  data: []
};

function reducer(state, action) {
  switch (action.type) {
    case 'FETCH_SUCCESS':
      return { loading: false, error: '', data: action.payload };
    case 'FETCH_ERROR':
      return { loading: false, error: 'Something went wrong!', data: [] };
    default:
      return state;
  }
}

function App() {
  const [state, dispatch] = useReducer(reducer, initialState);

  useEffect(() => {
    fetch('https://jsonplaceholder.typicode.com/users')
      .then(response => response.json())
      .then(data => dispatch({ type: 'FETCH_SUCCESS', payload: data }))
      .catch(() => dispatch({ type: 'FETCH_ERROR' }));
  }, []);

  return (
    <div className="App">
      <h1>useReducer Demo with Vite</h1>
      {state.loading ? (
        <p>Loading...</p>
      ) : state.error ? (
```

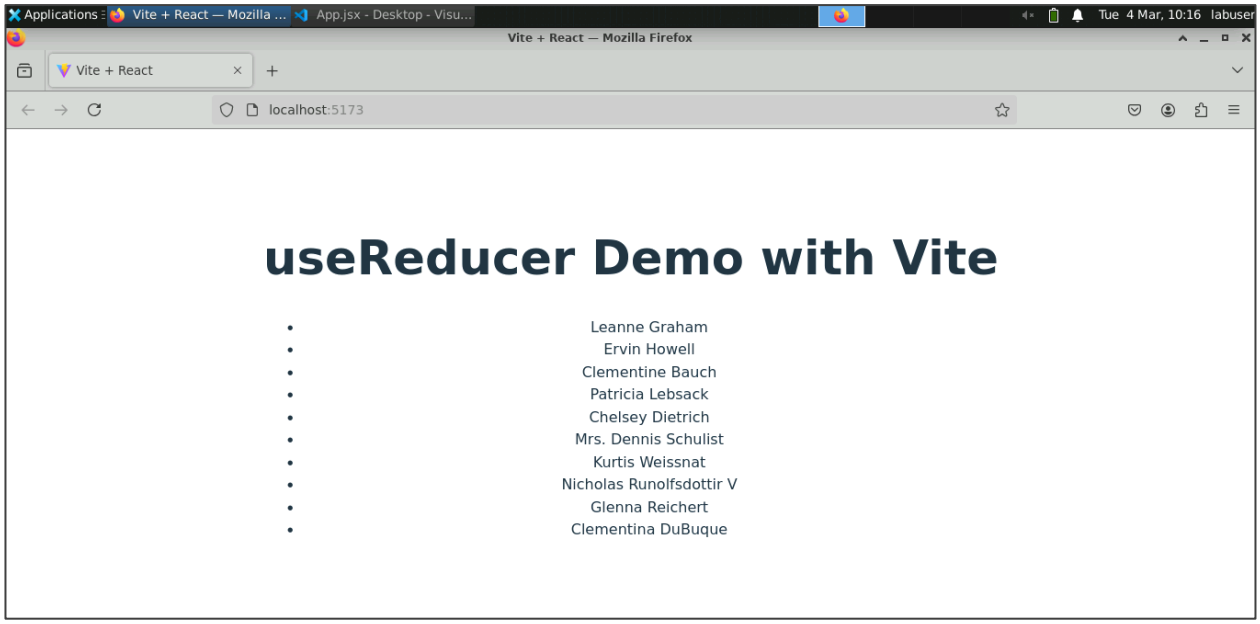
```
      <p>{state.error}</p>
    ) : (
      <ul>
        {state.data.map(user => (
          <li key={user.id}>{user.name}</li>
        ))}
      </ul>
    )}
  </div>
 );
}

export default App;
```



Note: The fetch function sends a request to **https://jsonplaceholder.typicode.com/users** and returns a Promise that resolves to give the response. The response is then parsed into JSON format.

Step 3: Run and verify the application

3.1 Navigate to the browser and refresh the webpage



Note: The application should display a list of usernames retrieved from the **JSONPlaceholder API.**

By following the above steps, you have set up a React app with Vite and used useReducer for efficient state management and data fetching. You can further enhance this application by improving error handling and refining the UI.